Ain Shams University
Faculty of Engineering
Electronics & Comm. Eng. Dept.

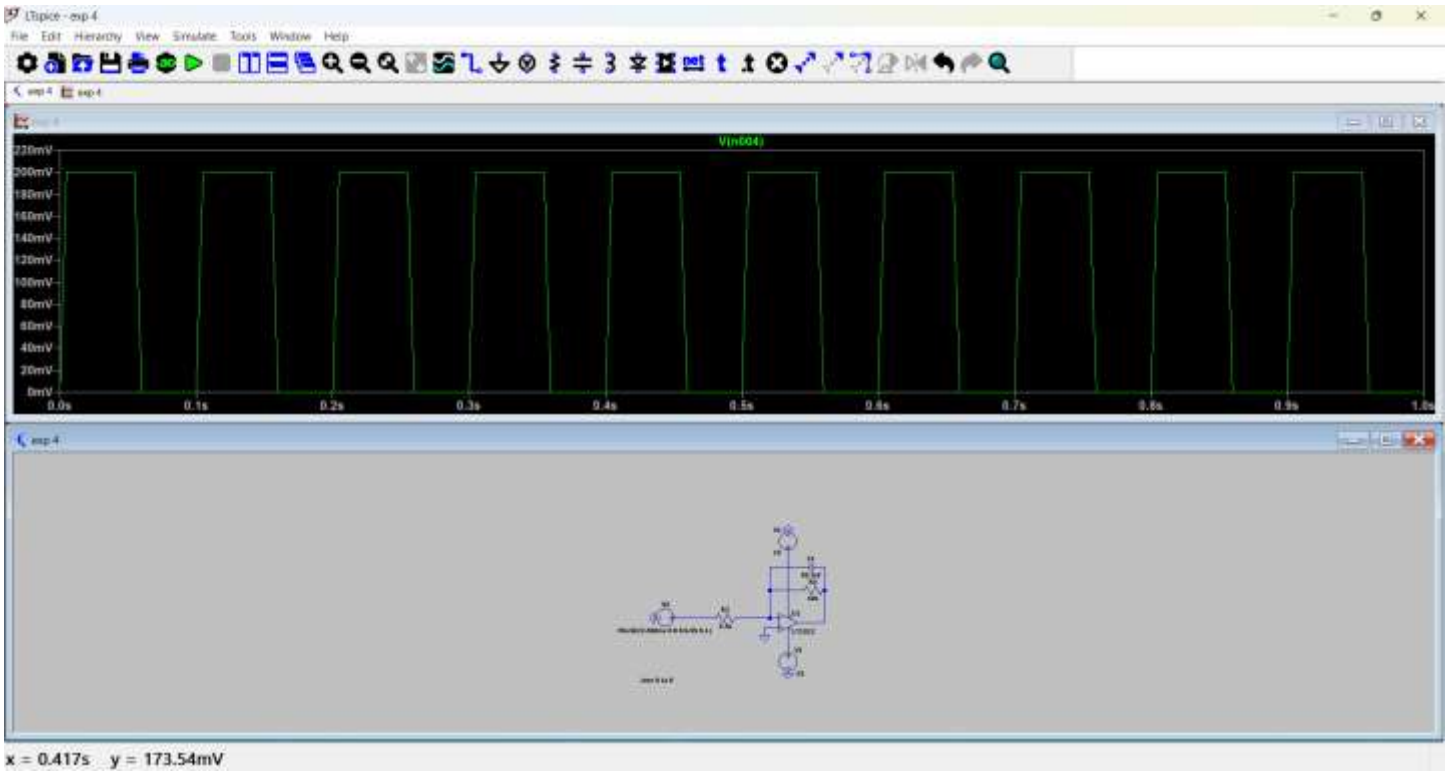


ECE211: Electronics
2nd Year Electrical Engineering
Fall 2024

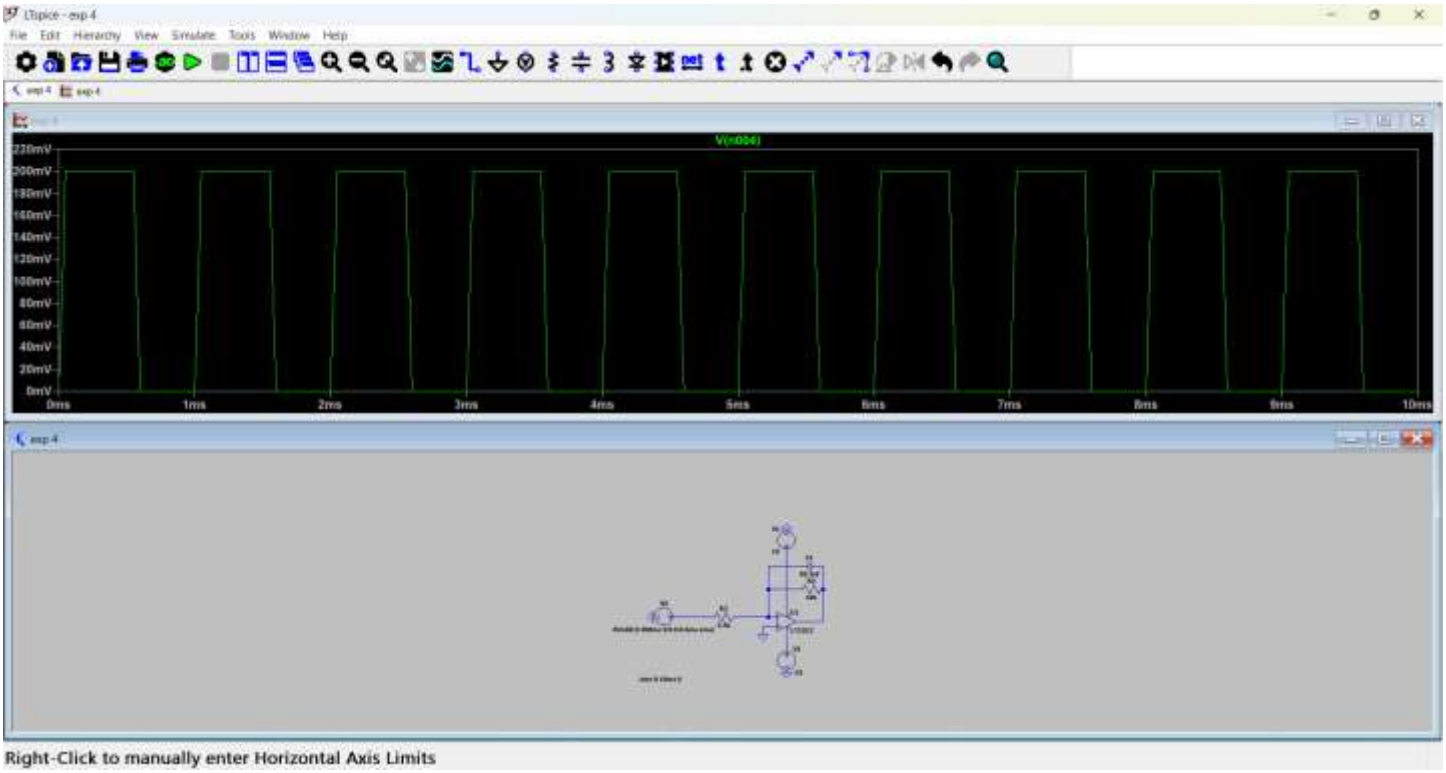
Experiment (Z) Report

Names	IDs	Department	Attendance slot	Group No.	Group Code Part A, B
Abdelrahman sherif	2300100	ECE	From 12:30 – 2:30 First Rotation	1	Part D
Second Name	ZZZZZZZ	ZZZ			
Third Name	ZZZZZZZ	ZZZ			

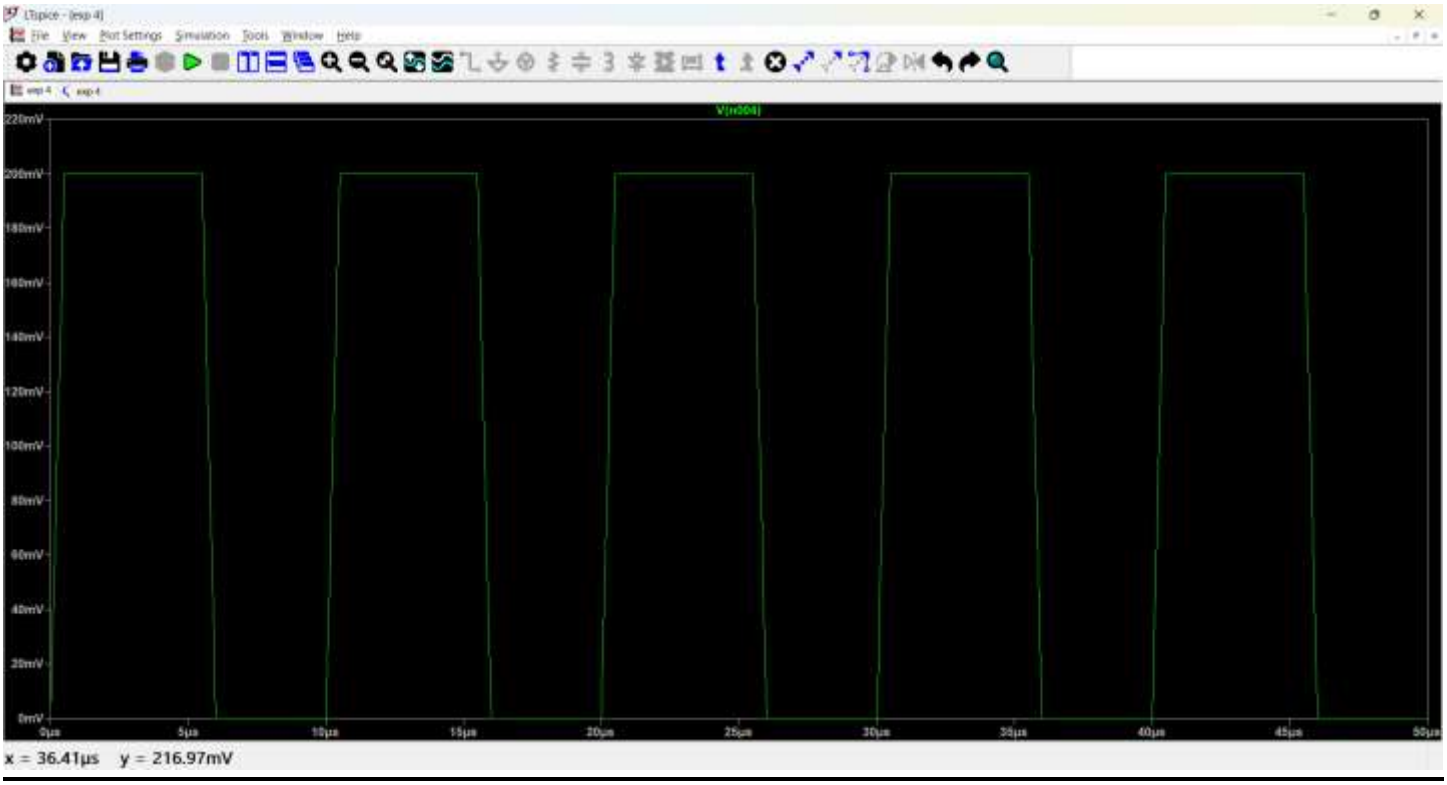
Part A:


LTspice - exp 4
File Edit Hierarchy View Simulate Tools Window Help
V(in004)
220mV
200mV
180mV
160mV
140mV
120mV
100mV
80mV
60mV
40mV
20mV
0mV
0.0s
0.1s
0.2s
0.3s
0.4s
0.5s
0.6s
0.7s
0.8s
0.9s
1.0s
exp 4
x = 0.417s y = 173.54mV

Part B:


LTspice - exp 4
File Edit Hierarchy View Simulate Tools Window Help
V(n004)
220mV
200mV
180mV
160mV
140mV
120mV
100mV
80mV
60mV
40mV
20mV
0mV
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms
exp 4
Right-Click to manually enter Horizontal Axis Limits

Part C:


LTspice - [exp 4]
File View Plot Settings Simulation Tools Window Help
V(in004)
220mV
200mV
180mV
160mV
140mV
120mV
100mV
80mV
60mV
40mV
20mV
0mV
0µs
5µs
10µs
15µs
20µs
25µs
30µs
35µs
40µs
45µs
50µs
x = 36.41µs y = 216.97mV

Part D:

The output response depends on the input frequency, as the integrator behaves similarly to a filter circuit. At low frequencies, the output appears nearly flat. At moderate frequencies, the output takes the form of a gradual charging and discharging curve. However, at high frequencies, the output starts to resemble the input signal due to reduced integration time.